

Title: EduPredict: Utilizing Machine Learning to Improve Student Retention

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1. Problem Identification In many educational institutions, student struggles are often identified too late—usually after final exams. This project aims to use behavioral data (attendance, study habits) to predict academic risk early in the semester.
2. Why it Matters Early intervention can significantly reduce dropout rates. By identifying "at-risk" students, teachers can provide targeted tutoring or resources, making education more equitable.
3. Approach Data: I used a dataset simulating student metrics like study hours and attendance.

Algorithm: I chose Random Forest because it handles non-linear relationships well and provides "Feature Importance," which tells us why a student might be struggling.

Evaluation: The model was evaluated using an Accuracy Score and a Confusion Matrix to ensure we aren't missing students who actually need help (False Negatives).

4. Challenges & Learning Challenge: Defining the "threshold" for a passing grade in a way that the model could learn effectively.

Learning: I learned that data quality (like attendance records) is often more important than the complexity of the algorithm itself.